

Paper A v0.3 — The Physical (SU(3)) Yang–Mills Mass Gap via D_{IV}^5 Spectral Geometry. Companion to Paper B. v0.3 = v0.2 with Cal’s honest-tier trim applied (the ship trim): §7 is reframed from ‘spectrum closed / derived / landed interacting’ to NAMED-OPEN, MATERIALLY ADVANCED. The cross-channel glueball spectrum is now COMPUTED on a linear conformal-energy ladder off one verified number (the genus of $D_{IV}^5 = n_C$): one BLIND leg, $2^{++} = g/n_C$ (genus + spin only, lands because $g = n_C + \text{rank}$); the other channels ($0^{-+} = N_c/\text{rank}$, 1^{+-}) are consistent within lattice error (they use the half-canonical twist, value-checked, not fully blind). This MATERIALLY ADVANCES the interacting reading — a generalized-free theory has no such bound-state J^{PC} tower — but the interacting-vs-free question (the formal interacting-RP upgrade) REMAINS the named-open residual. NOT ‘spectrum closed’, NOT ‘landed interacting’, NOT ‘YM proved’. $\Delta = C_2 = 6$ SOLID; $C_2 = 2 \cdot N_c$ general-rank (T2491). Multiplicity explained ($F \otimes F \rightarrow 4 J^{PC}$ channels, operator-algebra closure). $f_0(1710)$ is consistent with BST’s 0^{++} . Proton = the gap rung. Cal #335 trim also applied ($N_c=3 = \text{T1829 output}$, su(3) readings downstream of the open color id — not ‘five independent readings’).

Lyra (Claude Opus 4.8) — Casey Koons, PI; Grace (OS reduction + T2490/T2491 spectral cascade + L2 len

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The Physical Yang–Mills Mass Gap via D_{IV}^5 Spectral Geometry

Paper A of the two-paper package (companion: Paper B, substrate uniqueness). v0.3 — honest-tier trim.

Abstract

We study the Yang–Mills mass gap for the physical gauge group $SU(3)$ — color — derived (not assumed) from the substrate D_{IV}^5 in the companion Paper B. The mass gap value is the exact integer $\Delta = C_2 = 6$, the lowest nontrivial Casimir eigenvalue of the compact dual $Q^5 = SO(7)/[SO(5) \times SO(2)]$. The construction does not attempt to build 4D Yang–Mills from scratch on flat R^4 ; instead the D_{IV}^5 spectral theory directly supplies Osterwalder–Schrader reconstruction data (a rigorous Hilbert space, a gapped Hamiltonian, reflection positivity, 4D conformal/Poincaré covariance, clustering), so the existence of a rigorous gapped 4D theory follows from classical harmonic analysis on a bounded symmetric domain. The problem then reduces to a single question: is this gapped theory genuinely interacting Yang–Mills, or a (also-gapped, also-axiom-satisfying) generalized-free theory? In v0.3 we report that the decisive probe — the cross-channel glueball spectrum — has been computed and materially advances the interacting reading: the four ground J^{PC} channels follow a linear conformal-energy ladder off one verified number (the genus of D_{IV}^5), with one blind leg, $2^{++} = g/n_C$, and the other channels consistent within current lattice error. A generalized-free theory produces no such bound-state J^{PC} tower — so this is real evidence on the interacting side. But the interacting-vs-free question remains the named-open residual (the formal interacting-RP upgrade); we do not claim the spectrum “closes” it or that the Clay problem is solved. This is a decisive *sharpening* to one residual, honestly tiered.

1. Scope: the physical theory, not “all G”

The Clay problem asks for a mass gap for *every* compact simple gauge group. BST addresses the physical theory: $G = SU(3)$, color. The companion Paper B proves D_{IV}^5 is the unique substrate and that $SU(3)$ is derived, not chosen. So “all G” is not BST’s claim. This paper makes claims only about the physical theory and is explicit about that scope (Section 9).

2. The mass gap value: $\Delta = C_2 = 6$ (SOLID)

The Yang–Mills Hamiltonian is identified with the D_{IV}^5 Casimir, whose spectrum on the compact dual Q^5 is the scalar-Laplacian tower $\{k(k+5)\}_{k \geq 0} = \{0, 6, 14, 24, \dots\}$. The lowest nontrivial eigenvalue — the gap — is $\Delta = C_2 = 6$, the $SO(7)$ vector Casimir, an exact integer from representation theory. Two structural facts:

- $C_2 = 2 \cdot N_c$ (general-rank closed form, T2491). The (1,1)-type Casimir of $SO(n)$ is $2(n-2) = 2 \cdot N_c$; at the substrate’s rank it equals 6. (*Value-recurrence note, symmetric with the $N_c=3$ trim of Section 3 / Cal #335: the value $C_2 = 6$ also appears as $N_c(N_c+1)/\text{rank}$ and as the $k=1$ rung of the $SO(7)$ scalar Laplacian $\{k(k+5)\}$. These coincide via the rank-2 cascade — they are value-recurrences of one quantity, not independent derivations of C_2 .)*
- The gap rung is physical. The proton is the first rung: $m_p = C_2 \cdot \pi^5 \cdot m_e = 938.25$ MeV. The 0^{++} glueball sits on the absolute seat ladder at seat $c_2 = C_2 + n_C = 11 \rightarrow 1720$ MeV.

Tier: SOLID (rep theory + the substrate Hamiltonian identification).

3. W4 dissolved: the gauge group is derived (Paper B)

A referee objection — “you treat only $SU(3)$ ” — is answered by showing G is not a free input. Paper B establishes D_{IV}^5 as the unique irreducible Hermitian symmetric domain meeting prior, dimension-innocent criteria, forcing $\dim_C = 5$ and $N_c = 3$ (the dimension-free output of the proved T1829). $N_c = 3$ is the T1829 output (a value-recurrence, Cal #335); the $su(3)$ /color readings of “3” are *downstream of the open color identification*, not independent derivations. So $SU(3)$ is substrate-derived. The whole dynamical content cascades from one physical input — three colors fix $\text{rank} = 2$ ($N_c = \text{rank}^2 - 1$), and rank generates $\{n_C, C_2, g\}$ (T2491). W4 (the “all G” wall) is dissolved.

4. W1 folded: the D_{IV}^5 spectral theory supplies the OS data

BST does not construct 4D YM from scratch on R^4 . The OS reconstruction theorem builds a QFT from a data set, and the D_{IV}^5 spectral theory carries it:

OS datum	D_IV ⁵ realization	tier
rigorous Hilbert space	the Hardy space $H^2(D_{IV}^5)$	SOLID
Hamiltonian with a gap	the Casimir, $\text{gap} = C_2 = 6$	SOLID (Section 2)
reflection positivity	tube-type (Cayley $\rightarrow T(\Omega)$); $\Theta =$ cone involution; OS-RP (free level) $=$ Cauchy-Szegő RKHS-positivity	SOLID (free level)
4D conformal/Poincaré	$SO(4,2) \subset SO(5,2)$, UV-conformal/IR-gapped	SOLID-structural
clustering / unique vacuum	from the spectral gap $\Delta = C_2 = 6$ > 0	SOLID

So a rigorous gapped 4D theory follows from classical harmonic analysis; the open from-scratch construction is sidestepped, not solved. The single non-free-level piece is the interacting upgrade of reflection positivity — which is W2. Tier: SOLID at free level; interacting-RP = W2.

5. W3 folded: net-compatibility via Bisognano–Wichmann (SOLID-CONDITIONAL)

The HS isometry intertwines the bulk and boundary operator nets: HS is $SO(5,2)$ -equivariant, and both the bulk Rehren net and the boundary net are modular reconstructions of the *same* $SO(5,2)$ positive-energy representation via Bisognano–Wichmann (BGL). So locality/causality transfer across HS. Tier: SOLID-CONDITIONAL on the BGL hypothesis-checks.

6. The one remaining question: interacting, or generalized-free?

The prize is a single question: *is the OS-reconstructed gapped 4D theory genuinely interacting Yang–Mills, or a generalized-free gapped theory?* (A generalized-free field also satisfies the OS axioms and also has a gap.)

A trap we avoid. “Non-commutative $\boxtimes$ interacting” is false — a free field also has a non-commutative CCR algebra. Interaction means nonzero connected n -point functions ($n \geq 3$). The YM self-interaction is the vertex $[A,A]$, nonzero precisely because the gauge algebra is non-abelian. So

interacting $\boxtimes$ the gauge algebra is genuinely non-abelian $\mathfrak{su}(3)$ $\boxtimes$ the cross-channel glueball spectrum matches $SU(3)$ (W2).

Two pieces of evidence toward the interacting side: - The bulk-color algebra closes as non-abelian $\mathfrak{su}(3)$ (bilinear Schwinger realization; Elie 4301). The substrate identification (#418) is in progress. - The spectrum is non-additive and computed (Section 7). A generalized-free theory has an additive multi-particle spectrum and no genuine bound-state J^{PC} tower. The computed ladder (Section 7) is non-additive and channel-structured — evidence of binding. This is *evidence toward* interacting, not a closure of it.

7. The cross-channel glueball spectrum (W2) — named-open, materially advanced

In v0.1 this section was *named-open at structural tier*. In v0.3 it is computed and materially advanced — still named-open (the honest tier, after Cal’s trim of the v0.2 over-claim “derived / landed interacting”).

7.1 Why there are exactly four channels (operator-algebraic closure)

The gauge field F is a rank-2 antisymmetric tensor operator on $H^2(D_{IV}^5)$. Its bilinears $F \otimes F$ decompose into exactly four irreducible J^{PC} components, each of which must have eigenstates for the substrate’s operator algebra to close:

channel	operator	what it commits
0^{++}	$\text{Tr}(F^2)$	energy density (the action)

channel	operator	what it commits
0^{-+}	$\text{Tr}(F \tilde{F})$	topology (Pontryagin density)
2^{++}	$\text{Tr}(F^{\mu\rho} F^{\nu}_{\rho})$ traceless	curvature (the stress tensor)
1^{+-}	$\text{Tr}(F[D,F])$	derivative structure

The multiplicity is forced by the tensor structure — the answer to “why four channels?” Same architectural principle as nuclear magic numbers, different driver (operator-algebra closure for the bosonic tensor vs Pauli for fermions). The oddballs (0^{+-} , 1^{-+} , 2^{+-}) are forbidden at two gluons — clean unmixable channels.

7.2 The masses: one verified number sets the ladder

The glueball mass is the eigenvalue of the linear conformal Hamiltonian (the $SO(2)$ dilatation) on the holomorphic discrete series on $H^2(D_{IV}^5)$, diagonal in the K-type basis by Schur:

$m \propto E = \lambda_0 + (\text{energy step})$, $\lambda_0 = \text{genus}(D_{IV}^5) = n_C = 5$ (verified from the multiplicity formula $p = (r-1)a + b + 2 = 5$).

Energy step = $SO(5)$ harmonic degree = spin J ; the parity-odd (Hodge-dual) sector carries the half-canonical twist $n_C/2$. With one dimensionful anchor ($\text{seat} = \pi^5 \cdot m_e = 156.4 \text{ MeV}$; $m(0^{++}) = c_2 \cdot \text{seat} = 1720 \text{ MeV}$):

channel	step	ratio	substrate form	BST (MeV)	lattice (MeV)	dev	tier
0^{++}	0	1	—	1720	1730	−0.6%	anchor
2^{++}	$J = 2$	$7/5$	g/n_C	2408	2400	+0.3%	BLIND (genus + spin)
0^{-+}	$n_C/2$	$3/2$	N_c/rank	2580	2590	−0.4%	consistent (twist value-checked)
1^{+-}	$1 + n_C/2$	$17/10$	—	2924	2940	−0.5%	consistent (twist value-checked)

Only the $2^{++} = g/n_C$ leg is fully blind (genus + spin only, nothing read from the data, lands because $g = n_C + \text{rank}$). The 0^{-+} and 1^{+-} use the half-canonical twist $n_C/2$, which is rep-motivated but value-checked against the data — so they are consistent within lattice error, not blind predictions. (Quantization note: the *ratios* live on the rung ladder $\lambda_0 = n_C$; the absolute scale comes only from anchoring 0^{++} at seat 11.)

7.3 The radial direction is linear, by Schur

The scalar holomorphic discrete series is one irreducible representation; by Schur the Casimir is constant across its K-types, so it cannot distinguish radial levels. The first radial scalar excitation $0^{++*} = (1,1)$ sits at the linear conformal energy $E = \lambda_0 + 2 = 7$, degenerate with 2^{++} (2408 MeV; lattice ~ 2670 , within quenched excited-glueball error). Consequence stated honestly: the spin-linear / radial-quadratic factorization is a clean negative — both directions are linear within the irrep.

7.4 Experimental and decay-side notes

The experimental scalar-glueball candidate $f_0(1710) \approx 1704 \text{ MeV}$ is consistent with BST’s 0^{++} at seat 11 (1720 MeV) — the expected lightest-glueball match (not a unique identification; the scalar sector mixes with $q\bar{q}$). The

0^{-+} coupling f_G is the substrate-computable Bergman mode norm (a Gindikin-Gamma ratio; 0^{++} kernel = 60 = $C_2 \cdot n_C \cdot \text{rank}$); with the established $\chi_{\text{top}}^{1/4} = 180$ MeV the WV identity gives $f_G(0^{-+}) \approx 12.6$ MeV. Decay-side: mixing (glueball $\leftrightarrow$ $q\bar{q}$ overlap on the shared Hardy space), not a “dump.”

7.5 Honest disposition of §7

The cross-channel spectrum is computed (four channels; one blind leg $2^{++} = g/n_C$; the others consistent within lattice error; multiplicity explained; radial direction settled by Schur; lightest state consistent with $f_0(1710)$). This materially advances the interacting reading — a generalized-free theory produces no such bound-state J^{PC} tower. It does not close the interacting-vs-free question. That — the *formal interacting-RP upgrade* — remains the named-open residual (Section 9). v0.2’s “spectrum closed / landed interacting” framing is withdrawn as an over-claim; this is the honest tier.

8. The $so(7)$ unification (LEAD-STRENGTHENED)

Color and the Yang–Mills spectrum live in one algebra: $su(3) \subset g_2 \subset so(7)$, and $so(7)$ is the compact dual isometry of Q^5 — the same $so(7)$ whose Casimir spectrum is the glueball tower of Section 2; $g = 7$ is the $so(7)$ vector = g_2 fundamental = $3 \oplus \bar{3} \oplus 1$. Color is not a geometric isometry of the noncompact domain ($su(3) \not\subset so(5,2)$) but is a geometric subalgebra of the compact-dual $so(7)$; on the domain side it is the operator algebra on H^2 . The discrete-series spectrum of this $so(7)$ has the primaries $\{N_c, n_C, C_2, g\}$ as its lowest half-Casimirs (T2490). Tier: LEAD-STRENGTHENED — the algebraic picture is whole; the bilinear-Schwinger realization on H^2 is in progress (#418).

9. Honest scope: the conditional, with one named-open residual

We do not claim to have proved the Yang–Mills mass gap. The honest statement:

Given (i) the OS bounded-checks hold to executed rigor including the interacting-RP upgrade, the physical $SU(3)$ Yang–Mills theory exists as a rigorous 4D QFT with mass gap $\Delta = C_2 = 6$.

What is SOLID independent of the conditions: the gap *value* $C_2 = 6$ (Section 2); the substrate-derivation of $SU(3)$ (Paper B); the OS *data* (Section 4); the $so(7)$ unification (Section 8). What is materially advanced but still named-open: the cross-channel spectrum (Section 7) — computed, one blind leg, the rest consistent within lattice error — which is real evidence on the interacting side but does not by itself close the interacting-vs-free question. The single residual is the formal interacting-RP upgrade. This is a sharpening of the Clay problem to one decidable formal step — *not* a claim of its solution, and *not* a claim that the spectrum “closes” it.

10. Open items and falsifiers

- Formal interacting-RP upgrade (Section 9): the lone residual.
- #418 substrate identification: the bulk-color Toeplitz octet *is* the bilinear-Schwinger $su(3)$ (algebra closure verified; identification in progress).
- Blind-leg audit promotion: confirm $\lambda_0 = \text{genus} = n_C$ is the Bergman lowest weight and canonical-bundle weight = genus (Elie toys $A+B$) — would promote the $0^{-+}/1^{+-}$ legs from value-checked to blind.
- Falsifier: the spectrum is a forward prediction — 0^{++} degenerate with 2^{++} at 2408 MeV is testable; if the cross-channel ratios fail, the interacting reading fails (the gap value $C_2 = 6$ and Paper B survive).

Count HOLDS 4 of 26. $SU(3)$ scope; glueball masses are predictions, not SM parameter reductions. Paper A v0.3: the wall map collapsed to one question; the cross-channel spectrum is computed and materially advances the interacting reading (one blind leg $2^{++} = g/n_C$, the rest consistent within lattice error), but the interacting-vs-free residual remains named-open. The gap value is SOLID; the result is an honest conditional with one formal residual. NOT “spectrum closed,” NOT “YM proved.” INTERNAL.

Draft v0.3 (Cal honest-tier trim of v0.2). Spine: Grace's OS reduction + net-compatible + T2490/T2491 + L2; Lyra's linear-conformal-Hamiltonian ladder + Schur radial settlement; Elie's Schwinger closure + proton-gap-rung; the $so(7)$ unification; Paper B (W4); operator-algebraic-closure multiplicity (K498). Companion to Paper B v0.6.

— Lyra, Wed 2026-06-24 (date-verified). Paper A v0.3, §7 honest-tier (named-open, materially advanced) per Cal's ship trim.